Topic 4 — Biodiversity

Covers syllabus (click to expand)

Biodiversity | Levels of Biodiversity | Genetic Diversity (आनुवंशिक विविधता) | Species Diversity (प्रजाति विविधता) | Ecosystem (पारिस्थितिकी तंत्र) Diversity (पारिस्थितिक तंत्र विविधता) | Global Biodiversity | Indian Biodiversity | Importance (Values) of Biodiversity | Biodiversity Assessment | Biodiversity Registers | Biodiversity and Environment (पर्यावरण) | Biodiversity Loss | Biodiversity Conservation | Biodiversity Hotspots | Endemic Species (स्थानिक प्रजाति) | Endangered Species | Threatened Species (संकटग्रस्त प्रजातियां) | Extinct Species | Invasive Alien Species (आक्रामक विदेशी प्रजाति) | Keystone Species (कीस्टोन प्रजाति) | Flagship Species (प्रतीक (फ्लैगशिप) प्रजाति) | Umbrella Species (छत्र (अंब्रेला) प्रजाति) | Indicator Species (जैव संकेतक प्रजाति)

Sources baked in: NCERT Class 12 Biology Ch 15–16, CBD/Nagoya/Cartagena, Biological Diversity Act 2002, IUCN Red List, Conservation International hotspots, Ramsar (~100 sites; UP 12), UPPCS/UPSC PYQs 2018–2025 **Weight:** ★★★ Very High — definitions, hotspots, IUCN, loss drivers, conservation, Ramsar **Last verified:** September 2026

Current Affairs (this topic)

- Track new Ramsar (रामसर) and IUCN (आईयूसीएन) Red List updates against the Must-Score baselines (India Ramsar **~100**; UP **12**).
- Invasive-species and hotspot stems recycle yearly — keep Western Ghats (पश्चिमी घाट) / Indo-Burma trap pairs stable.

4.1 Biodiversity – Meaning

Identity: CBD Art 2 | Three nested levels | Rosen coined · Wilson popularised

- **Biodiversity** is the variety and variability of life at **genetic**, **species**, and **ecosystem** (पारिस्थितिकी तंत्र) levels in a given area — the diverse range of plants and animals in an ecosystem.
- The **Convention on Biological Diversity (CBD, 1992)** defines it as variability among living organisms from all sources, including terrestrial, marine, and aquatic ecosystems.
- **Walter G. Rosen (1985)** coined the word *biodiversity* (from “biological diversity”) for a National Research Council seminar title. **E.O. Wilson** popularised the term thereafter.
- Biodiversity is **not** just a species count. Genes within populations and the variety of habitats also matter.
- The most significant aspect of biodiversity is **maintenance of the ecosystem**, ahead of food, drugs, or industrial use alone.
- **International Day for Biological Diversity** is **22 May** (UN moved it from the older **29 December** CBD entry-into-force date). The UN declared **2011–2020** the **Decade on Biodiversity**.

- India's first **National Centre for Marine Biodiversity (NCMB)** is at **Jamnagar (Gujarat)**.

> **Logic:** "Biodiversity = only number of species" is false — all three CBD/NCERT levels are required. Rosen coined; Wilson popularised — do not reverse.

4.2 Levels of Biodiversity

Identity: Genetic | Species | Ecosystem | Whittaker α–β–γ

Genetic diversity

- **Genetic diversity** is variation in genes and alleles **within** a species.
- It gives populations resistance to disease and ability to adapt to environmental change.
- Crop **landraces** and wild relatives of rice, mango, and pulses are India's genetic wealth.

Species diversity

- **Species diversity** depends on **species richness** (how many species) and **evenness** (how evenly individuals are spread).
- **R.H. Whittaker (1960)** advanced the **alpha–beta–gamma** scale for species diversity.
- **Alpha diversity** (अल्फा विविधता) is the **mean species diversity in habitats at a local scale**.
- **Beta diversity** (बीटा विविधता) is the comparison / change of species **between** ecosystems.
- **Gamma diversity** is overall diversity across different ecosystems in a **region**.
- Species diversity is highest in the **tropics** and generally **decreases toward the poles** (latitudinal diversity gradient). Along mountains, diversity is normally greater at **lower altitudes** than at higher ones.

Ecosystem diversity

- **Ecosystem diversity** is the variety of habitats and ecosystems in a region such as forests, wetlands, grasslands, and deserts.
- India spans **10 biogeographic zones**, so ecosystem diversity is high despite moderate (नरम दल) land area.
- On land, **tropical rainforest** (about **23.5°N–23.5°S**) is the richest biome — often called the **optimum biome**. In the sea, **coral reefs** rank among the richest systems; both can be accepted when a stem asks for relatively higher diversity.

> **Logic:** "Species diversity increases from equator toward poles" is **false**. Alpha = local mean; beta = between ecosystems; gamma = regional total.

4.3 Global & Indian Biodiversity

Identity: ~8.7M species estimated | 17 megadiverse nations | India 2.4% land, 7–8% species

- About **8.7 million** species may exist on Earth (Mora et al., 2011). Only about **1.8 million** are described so far.
- The **tropics** hold the greatest species diversity because of long evolutionary history (इतिहास), stable climate, and high productivity.
- **17 megadiverse countries** together hold most of the world's species. **India** is one of them.
- **Russia is not** on the standard megadiverse list of 17.

Region	Megadiverse countries (17)
South America	Brazil, Colombia, Ecuador, Peru, Venezuela
North / Central America	Mexico, United States
Asia	China, India, Malaysia, Indonesia, Philippines
Africa	Democratic Republic of Congo, South Africa, Madagascar
Oceania	Australia, Papua New Guinea

- India covers about **2.4%** of the world's land but holds roughly **7–8%** of recorded global species.
- India has about **45,000** plant species and **91,000+** animal species (NCERT order of magnitude).

Group	India's rough share of global species
Mammals	**~7.6%**
Birds	**~12.6%**
Reptiles	**~6.2%**
Amphibians	**~4.4%**
Indian endemism centre	Note
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Western Ghats (पश्चिमी घाट)	Richest among common Indian options
Northeast India	High endemism belt
Himalaya (हिमालय)	Richness from confluence of biogeographic zones
Andaman & Nicobar Islands	Island endemism

- Among common Indian options, the **Western Ghats** is the most biodiversity-rich region.
- **Silent Valley (Kerala, Palakkad)** is remnant moist evergreen rainforest of the Western Ghats.
- Silent Valley was declared a **National Park in 1984** (inaugurated **1985**).
- The **Silent Valley Project** was the Kerala hydel controversy linked to this forest.
- Among named Indian valleys in older stems, **Silent Valley** (साइलेंट वैली) is the richest.
- The **Valley of Flowers** lies in **Chamoli, Uttarakhand**.
- Himalayan species richness is best explained as a **confluence of different biogeographic zones**, not rainfall alone.

> **Logic:** Western Ghats beats Gangetic Plain, Trans-Himalaya, and Central India for species richness. Silent Valley = Kerala, not Uttarakhand.

PYQ — UPPCS Prelims 2019, Q139

The most biodiversity rich area in India is

A. Gangetic plain

B. Trans Himalayas

C. Western Ghats

D. Central India

Show answer

Logic: Stem tests India's richest mainland biodiversity region among standard distractors — Gangetic Plain, Trans-Himalaya, and Central India are species-poorer than the Western Ghats hotspot. **Ans: C.** The **Western Ghats** hold the highest species richness among these options. Gangetic Plain, Trans-Himalaya, and Central India are classic weaker distractors. Silent Valley (Kerala) sits inside this Western Ghats richness belt — do not confuse it with Valley of Flowers (Chamoli, Uttarakhand).

4.4 Importance & Biodiversity–Environment Link

Identity: Direct + indirect + option values | Ecosystem maintenance first

- The most significant aspect of biodiversity is **maintenance of the ecosystem**.
- **Direct values** include food, fuel, fibre, medicine, and timber taken from nature.
- **Indirect values** include soil formation, prevention of soil erosion, recycling of waste, pollination of crops, climate regulation, and flood control.
- **Ethical and aesthetic values** cover cultural, spiritual, and recreational benefits.
- **Option value** is the potential future use of species and genes not yet exploited.
- Higher biodiversity often means **greater ecosystem stability** and resilience after disturbance.
- Biodiverse systems recover faster from drought (सूखा), pests, and pollution shocks.
- Maximum butterfly species occur in **tropical** countries because butterflies **cannot tolerate low temperatures**.

> **Logic:** Food/drugs matter, but stems asking the “most significant aspect” key **ecosystem maintenance**. Soil formation + pollination + waste recycling are classic multi-statement true sets.

4.5 Biodiversity Assessment & Registers

Identity: IUCN Red List | BDA 2002 | PBR + BMC

IUCN Red List categories

Category	Meaning
LC	Least Concern
NT	Near Threatened
VU	Vulnerable
EN	Endangered
CR	Critically Endangered
EW	Extinct in the Wild
EX	Extinct

- **Threatened species** = **VU + EN + CR**.
- **Endangered** means **EN only**, not all threatened categories.
- **DD** (Data Deficient) and **NE** (Not Evaluated) mean insufficient data for a full assessment.
- **IUCN** (आईयूसीएन) is headquartered in **Gland, Switzerland**. It assesses species status but does **not** regulate international trade (**CITES** (साइट्स) does that).
- India's **Red Data Book** lists threatened flora (वनस्पति) (BSI) and fauna (प्राणीजात) (ZSI) separately.
- The **Living Planet Index (LPI)** tracks global vertebrate population trends. **WWF** publishes it.
- **IPBES** (Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services) is headquartered in **Bonn, Germany**. It gives biodiversity assessment reports similar to IPCC (आईपीसीसी) for climate.

People's Biodiversity Register (India)

- The **Biological Diversity Act, 2002** implements the CBD in India.
- The **National Biodiversity Authority (NBA)** (राष्ट्रीय जैव विविधता प्राधिकरण) is in **Chennai**.
- **State Biodiversity Boards (SBB)** and **Biodiversity Management Committees (BMC)** operate at state and local levels.
- A **People's Biodiversity Register (PBR)** (जन जैव विविधता रजिस्टर) documents local biodiversity and traditional knowledge at the panchayat level.
- The **Nagoya Protocol (2010)** sets rules for **Access and Benefit Sharing (ABS)** of genetic resources under the CBD.
- **Biodiversity Management Committees (BMCs)** are central to Nagoya-style ABS; they may levy **collection fees** on access to biological resources in their jurisdiction.
- **CEBPOL** (Centre for Biodiversity Policy and Law), with Norway, sits with the NBA in **Chennai**.

> **Logic:** NBA HQ is **Chennai**, not New Delhi. **Threatened ≠ Endangered**. NBA does **not** directly run GEAC genetic-modification supervision.

4.6 Biodiversity Loss

Identity: Habitat destruction #1 | HIPPO | Jhum · invasives · trophic levels

- **Habitat loss and fragmentation** is the **leading direct cause** of biodiversity loss worldwide.
- **Habitat fragmentation** splits large habitats into small patches. Species face edge effects, inbreeding, and local extinction.
- **Over-exploitation** through poaching, overfishing, and unsustainable harvesting removes species faster than they reproduce.
- **Invasive alien species** outcompete or prey on natives. **Water hyacinth** choking UP ponds and **Prosopis juliflora** wiping native flora are classic examples.
- **Pollution** kills pollinators, poisons aquatic life, and degrades breeding habitats.
- **Climate change (जलवायु परिवर्तन) / global warming** shifts ranges faster than many species can move or adapt.
- **HIPPO** summarises major drivers: **H**abitat loss, **I**nvasive species, **P**ollution, **P**opulation pressure, **O**ver-exploitation.
- **Shifting cultivation (jhum)** in eastern and northeastern India degrades habitats when practised intensively.
- India's bird populations shrink mainly from **habitat loss** and **pesticides / chemical fertilisers / mosquito repellents** (e.g. **DDT** eggshell thinning) — not from large-scale use of bio-pesticides.
- **Promotion of vegetarianism** is **not** a biodiversity threat. A **fewer number of trophic levels** indicates dominance of few organisms and **lowers** diversity — so it does **not** increase biodiversity.
- Afforestation of barren land is **not** a cause of biodiversity loss; road expansion, urbanisation, and agricultural extension are.
- Current extinction rates may be **100–1,000 times** the natural background rate.
- Scientists describe the present era as a **sixth mass extinction** driven mainly by human activity.
- The **species–area relationship** states that larger habitats generally support more species.
- **IPBES (2019)** reported that about **1 million** animal and plant species face extinction risk if trends continue.
- India aims for **Land Degradation Neutrality (LDN) (भूमि क्षरण तटस्थता) by 2030**.

> **Logic:** Destruction of **natural habitat** is the #1 loss factor. Pollution or climate change is not #1 in standard Prelims keys. Vegetarianism is a classic false threat option.

PYQ — UPPCS Prelims 2023, Q100

Which of the following is the **most important factor** responsible for the loss of biodiversity?

A. Genetic assimilation

B. Control of predators

C. Destruction of natural habitat

D. Pest control

Show answer

Ans: C

4.7 Biodiversity Conservation

Identity: In-situ preferred | Ex-situ backup | Biosphere + culture | CBD 3 objectives | 30×30

- **In-situ conservation** (स्व-स्थाने संरक्षण) protects species in their **natural habitat**. National parks, wildlife sanctuaries, biosphere reserves, reserved/protected forests, nature reserves, and **sacred groves** (पवित्र उपवन) are examples.
- **Ex-situ conservation** (बाह्य-स्थाने संरक्षण) protects species **outside** natural habitat in zoos, aquaria, botanical gardens, seed banks, and captive breeding centres.
- Cryo-banks store seeds, pollen, tissue, or embryos in **liquid nitrogen** for long-term ex-situ storage.
- **In-situ is preferred** because whole ecosystems and natural selection processes stay intact.
- **Ex-situ** is a **supplementary** insurance for critically endangered species.
- **Ex-situ** examples include the **National Bureau of Plant Genetic Resources (NBPGR)** seed bank in New Delhi, zoos, and botanical gardens.
- The **most important strategy for biodiversity together with traditional human life / cultural diversity** is the establishment of **biosphere reserves** (UNESCO MAB logic).
- India has **18** biosphere reserves; a majority are on the UNESCO World Network of Biosphere Reserves. **Gulf of Mannar** is often cited as India's richest **marine** biodiversity biosphere mosaic (21 islands with reefs, seagrass, mangroves).
- India's coral reefs (~**5,790 km²**) lie mainly in **Lakshadweep (atolls)**, **Gulf of Mannar**, **Gulf of Kachchh**, and **Andaman & Nicobar**. **Sundarbans** (सुंदरबन) is mangrove-dominated, **not** a coral-reef region. **Coral bleaching** is driven mainly by **rise in sea-water temperature / global warming**.
- The **Wildlife (Protection) Act, 1972** gives maximum protection to **Schedule I** species.
- The **CBD (1992)** has three objectives: **conserve** biodiversity, **use it sustainably**, and share benefits **fairly and equitably**.
- The **Cartagena Protocol on Biosafety** covers safe handling of **living modified organisms (LMOs)**; India implements it through **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय).
- The **Kunming–Montreal Global Biodiversity Framework (2022)** targets protecting **30%** of land and oceans by **2030** (30×30).
- CBD COP chronology: **COP-10 Nagoya (2010)**, **COP-11 Hyderabad (2012)**, COP-12 Korea, COP-13 Cancun, COP-14 Egypt (मिस्र), COP-15 Kunming–Montreal.
- **Project Tiger (प्रोजेक्ट टाइगर) (1973)** (प्रोजेक्ट टाइगर) uses the **umbrella/flagship** approach to protect forest ecosystems.

- **BirdLife International** is a global partnership of conservation organisations that identifies **Important Bird and Biodiversity Areas** — it did **not** invent biodiversity hotspots (**Norman Myers** did).
- **TRAFFIC** (Trade Records Analysis of Flora and Fauna in Commerce, **1976**) is a **WWF–IUCN** alliance monitoring wildlife trade — **not** a UNEP bureau.
- **TEEB** (The Economics of Ecosystems and Biodiversity) makes nature's economic values visible for decision-makers.
- **Wetlands International** is an independent global NGO (HQ **Netherlands**) working on wetlands — not an intergovernmental Ramsar organ.
- **UN-REDD+** (FAO–UNDP (यूएनडीपी)–UNEP) can protect biodiversity, forest resilience, and support poverty reduction when well designed.

> **Logic:** Botanical garden = ex-situ, not in-situ. Biosphere reserve wins stems that join biodiversity with traditional human life. Cryo gas = **nitrogen**.

PYQ — UPPCS Prelims 2023, Q118

In National Parks, genetic diversity is maintained through—

A. In-situ conservation

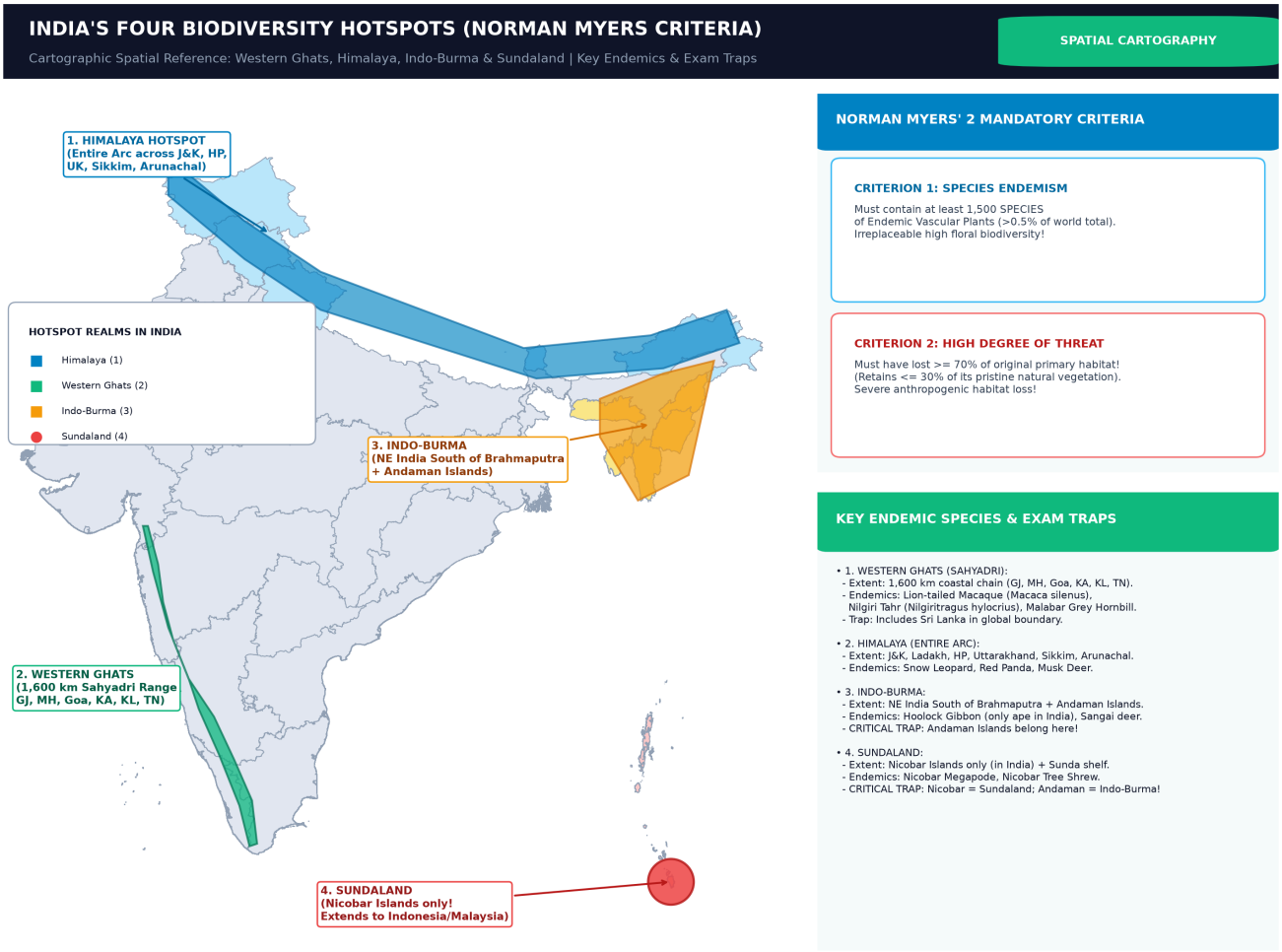
B. Ex-situ conservation

C. Gene pool

D. None of the above

Show answer

Ans: A



4.8 Biodiversity Hotspots

Identity: Myers 1988 | ≥1500 endemic plants + ≥70% habitat lost | India has 4 of 36

- A **biodiversity hotspot** (जैव विविधता हॉटस्पॉट) is a region with **high endemism** and **severe habitat loss**.
- **Norman Myers (1988)** set the hotspot concept. Conservation International later updated the global list to **36 hotspots** (about **2.5%** of Earth's land; **>50%** of endemic plant species). The **36th** hotspot added via CEPF (2016) is the **North American Coastal Plain**.
- **Two formal criteria:** at least **1,500 endemic vascular plant species** and at least **70% loss** of original primary vegetation. Recognition (मान्यता) also stresses **species richness, endemism, and threat perception** — not vegetation density or ethnobotany alone.
- Hotspots are **not** confined to the tropics; temperate examples include the California Floristic Province and Japan.
- **India has four hotspots:**

Hotspot	Indian region	UP link
Himalaya (often called Eastern Himalaya in older stems)	Himalaya + Terai belt; Sikkim is a botanist's paradise here	Dudhwa, Pilibhit Terai
Indo-Burma / Indo-Myanmar	Northeast India + **Andaman Islands**	—
Western Ghats & Sri Lanka	West coast mountains	—

Hotspot	Indian region	UP link
Sundaland	**Nicobar Islands** (not Andaman)	Not mainland

- **Eastern Ghats, Vindhyas, Aravallis, Gangetic Plain, Naro Hills, and Silent Valley alone** are **not** global hotspots (Silent Valley sits inside the Western Ghats hotspot).
- The **Gangetic Plain** is fertile and species-rich in places but is **not** a global biodiversity hotspot.

> **Logic:** Hotspot needs **endemism + threat**. Andaman ≠ Nicobar for hotspot mapping. Western Himalayas / Eastern Ghats are classic wrong options.

PYQ — UPPCS Prelims 2024, Q33

Consider the following statements regarding biodiversity hotspots:

1. They have high species endemism.
2. They have experienced significant habitat loss.

A. Only 1

B. Only 2

C. Both 1 and 2

D. Neither 1 nor 2

Show answer

Ans: C

4.9 Species Categories

Identity: Nine roles | IUCN status vs ecological role

Category	One-line fact	India example
Endemic	Found **nowhere else** in the world	Asiatic lion (सिंह) (Gir (गीर)), lion-tailed macaque (Western Ghats)
Endangered	IUCN **EN** — very high extinction risk	Gangetic dolphin, Indian vulture
Threatened	**VU + EN + CR** together	Broader than endangered alone
Extinct	No surviving individuals (**EX**) or wild population (**EW**)	Cheetah in India (1952 wild extinction)
Invasive alien	Non-native + ecological or economic harm	Water hyacinth, lantana, Parthenium
Keystone	Disproportionately large ecosystem impact if removed	Sea otter (kelp forest), tiger (apex regulation)

Category	One-line fact	India example
Flagship	Charismatic species that mobilises public support	Tiger, elephant, one-horned rhino
Umbrella	Protecting its large range saves many other species	Tiger reserve network
Indicator	Reflects environmental quality or change	**Lichens** (air pollution), frog (wetland health)

- An **endemic species** is found only in one defined geographic area. The Asiatic lion occurs only in Gir, Gujarat.
- An **endangered species** faces a very high risk of extinction in the wild under IUCN category **EN**.
- A **threatened species** is any species in **VU**, **EN**, or **CR** combined. It is a wider group than endangered alone.
- An **extinct species (EX)** has no surviving individuals. **Extinct in the Wild (EW)** survives only in captivity.
- An **invasive alien species** is non-native and harms native ecosystems or the economy. Water hyacinth chokes UP wetlands.
- A **keystone species** has ecosystem impact far greater than its numbers suggest. Robert **Paine (1969)** showed this in starfish experiments.
- A **flagship species** is a charismatic animal used to raise public support and funds for conservation, such as the tiger on campaign logos.
- An **umbrella species** needs a very large home range, so protecting it also protects many other species in the same habitat.
- An **indicator species** shows habitat or pollution conditions. Lichens fail in polluted air and signal air quality decline.
- Older definitional stems treat a species as **extinct** if not seen in its natural habitat for **50 years**.
- **Diclofenac sodium** (veterinary NSAID) caused India's vulture crash via contaminated carcasses.
- The **dodo** of Mauritius helped disperse **Tambalacoque** seeds; plant failure followed bird extinction.
- **Indian flying fox** is a **bat** (fruit bat), not a bird.
- **Dugong** is a **herbivorous marine mammal** ("sea cow"), Schedule I, mainly Gulf of Mannar / Palk / parts of Gujarat and A&N — **not** the entire Indian coast.
- **Indian wild ass** lives in the sandy saline (लवणीय) **Rann of Kachchh**; threat is habitat destruction, not predators.
- **Lion-tailed macaque** is endemic to the **Western Ghats** (Kerala, Karnataka, Tamil Nadu).
- **Cheetah** was declared extinct in India in **1952** (reintroduction is a later policy story).
- **Red panda** occurs in the eastern Himalayan belt (Sikkim, West Bengal, Meghalaya (मेघालय), Arunachal Pradesh, etc.) — **not** western Himalayas only. **Slow loris** lives in dense forests of the Northeast.
- **Shrew and tapir** are Himalayan-range animals, **not** Malabar Western Ghats.
- **Salt-water crocodile** occurs in the Andaman and Nicobar Islands.
- **Sea buckthorn** grows in cold deserts of **Ladakh** (लद्दाख) / Lahaul–Spiti and related Himalayan tracts; it checks erosion and is nutritious — not mainly a biodiesel/timber crop.

- **Prosopis juliflora** is an invasive exotic that reduces native biodiversity where it spreads.
- **Sequoia (Redwood)** of California is the tallest known living tree. Deep sea has **minimum** annual biomass production among common biome options.
- Hibernation occurs in **bats, bears, and many rodents**. Author of *Tomorrow's Biodiversity* is **Vandana Shiva**.
- **Bhitarkanika** (Odisha) sits where **Brahmani** (South Koel in Jharkhand) and **Vaitarni** meet near the Bay of Bengal.

Ramsar wetlands (biodiversity overlap)

- **Ramsar Convention (1971, Iran)** conserves wetlands of international importance under **Wise Use**. **World Wetlands Day (विश्व आर्द्रभूमि दिवस) = 2 February**.
- India has about **100** Ramsar sites. State counts often tested: **Tamil Nadu 16** (highest), **Uttar Pradesh 10**, **Punjab 6**. Bihar's first is **Kabartal / Kanwar Lake (Begusarai)**.
- **Montreux Record** lists Ramsar sites where ecological character has changed or is likely to change from human interference.
- Wetlands act as "kidneys" mainly by aquatic plants absorbing **heavy metals and excess nutrients**; they also cycle nutrients, hold sediment, and buffer floods.
- **Sambhar Lake (Rajasthan)** is India's largest inland saline wetland. Wetland Rules **2010/2017** and catchment coverage are Indian law — not every wetland worldwide is a mandatory Ramsar obligation.

> **Logic:** Lichens are pollution indicators because they **fail** in polluted air. Flying fox = bat. Dugong = mammal. Sundarbans ≠ coral reef. Ramsar count and TN/UP ranks move with notifications — verify live MoEFCC list before the paper year.
